

Machine Learning Model Evaluation — Study Map

One page, two notebooks: where every topic lives, and in which order to read it

ML_Model_Evaluation_Simple.ipynb (beginner) ⇄ ML_Model_Evaluation.ipynb (full course)

How to use this map

The 17 Simple sections are the core curriculum, in order. Each row shows where the same topic is treated in depth — jump there whenever you want more, or follow the ‘go deeper’ / ‘back to basics’ cards at the end of every section.

The 17 Simple sections → where each is covered in depth

#	Simple section	Deep counterpart	What the deep version adds
1	Introduction to Model Evaluation	1 · Introduction to Model Evaluation (+ Core Teaching Principle)	the three questions of evaluation, overfitting vs generalization
2	The Confusion Matrix	3 · The Confusion Matrix	deeper error-cost framing, more examples
3	Accuracy	4 · Accuracy	the accuracy trap in depth, when accuracy is the wrong metric
4	Precision	5 · Precision	more false-alarm cases, precision vs threshold
5	Recall (Sensitivity)	6 · Recall / Sensitivity	more missed-positive scenarios, multiclass recall
6	Specificity	7 · Specificity	precision vs specificity, the negative class in depth
7	F1 Score	8 · F1 Score	$F\beta$ / weighted F1, when F1 misleads
8	The Precision-Recall Trade-off	9 · The Precision-Recall Trade-off	cost-based threshold choice, the PR curve
9	ROC Curve and AUC	10 · The ROC Curve · 11 · AUC · 12 · ROC-AUC vs PR	what AUC really measures, PR vs ROC on imbalance
10	The Classification Report	14 · The Classification Report	macro vs weighted averages, multiclass use
11	Imbalanced Datasets	15 · Imbalanced Datasets	resampling, class weights, threshold tuning
12	Cross-Validation	17 · Cross-Validation	K-Fold in depth, common CV mistakes
13	Regression Metrics	20 · Regression Metrics	MAPE, choosing among regression metrics
14	Mini Case Study: Loan Default	21 · End-to-End Case Study · 22 · Business Decision · 23 · Cost-Sensitive	three models, the cost-matrix decision rule
15	Cheat Sheet	24 · Choosing the Right Metric · 27 · Final Cheat Sheet	the selection guide and the complete summary
16	Practice Exercises	28 · Practice Exercises	15 exercises with full solutions
17	Final Quiz	29 · Final Quiz	25 questions plus the final workflow

Deep-only topics — not in the Simple notebook

Meet these only in the full course; they extend the ideas above (reading order: follow the numbered sections).

2 · Classification vs Regression — framing only; light in Simple	18 · Data Leakage — When Your Model Cheats — leaked features inflate metrics
12 · ROC-AUC vs Precision-Recall — why PR beats ROC for rare events	19 · Time-Series Evaluation — forward-chaining, grouped splits
13 · Probability Calibration — rank ≠ calibration; reliability diagrams	25 · Common Mistakes — the pitfalls, catalogued
16 · Multiclass Evaluation — per-class metrics, averaging	26 · Beginner vs Expert Summary — the same ideas at two levels

Recommended reading order

Beginner: Simple 1 → 2 → ... → 17 straight through (≈ 1-2 hours, no prerequisites)
Deep: full course 1 → 2 → ... → 29 in order · Hybrid: start Simple, jump to depth via the cards